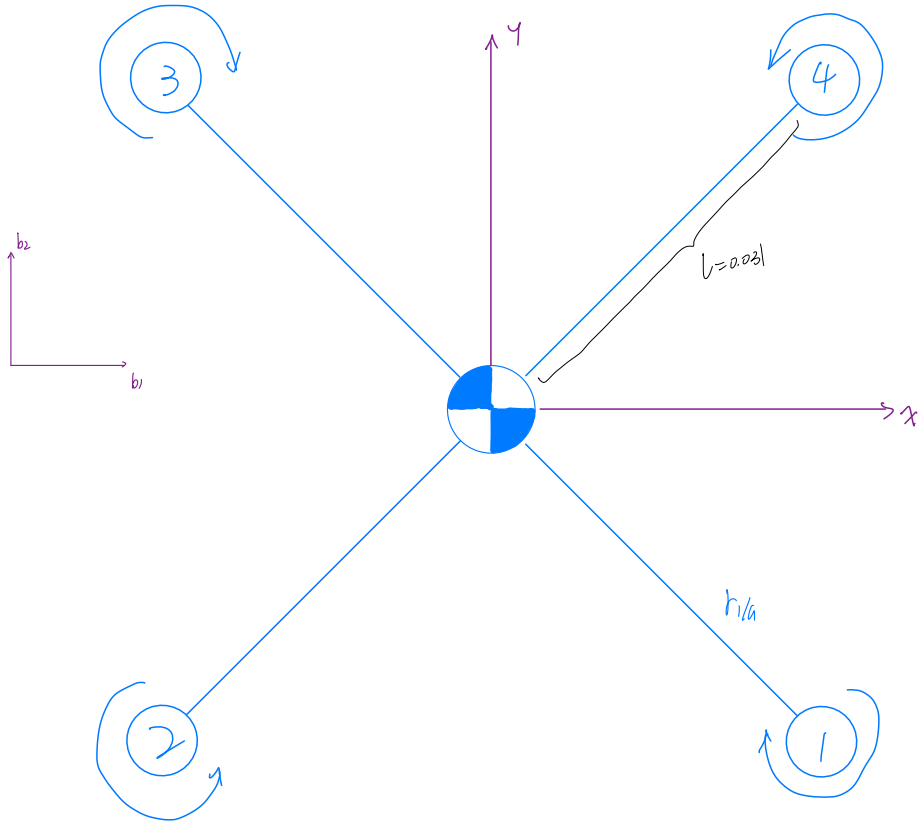


each propeller exerts a force (thrust) $F_i = k_T \omega_i^2$ in b_3 direction
a torque $\tau_i = k_M \omega_i^2$ in $\pm b_3$ direction
} ω_i is the angular velocity of each propeller

$$ {}^{B}\vec{F}_{ext} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ k_T & k_T & k_T & k_T \end{bmatrix} \begin{bmatrix} \omega_1^2 \\ \omega_2^2 \\ \omega_3^2 \\ \omega_4^2 \end{bmatrix} + \vec{a} \begin{bmatrix} 0 \\ 0 \\ -mg \end{bmatrix} $$

$$ \left(\vec{M}_{a,ext}\right)_{/B} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ k_M & -k_M & k_M & -k_M \end{bmatrix} \begin{bmatrix} \omega_1^2 \\ \omega_2^2 \\ \omega_3^2 \\ \omega_4^2 \end{bmatrix} + \sum_i {}^{B}\vec{r}_{i/4} \times \begin{bmatrix} 0 \\ 0 \\ k_T \end{bmatrix} \omega_i^2 $$

· Newton Euler Equation

Euler 1st law (translation): $\frac{d}{dt}{}^I\vec{P}_{a/o} = m_a \cdot {}^I\vec{a}_{a/o} = \vec{F}_{exe}$

$${}^I\vec{P}_{a/o} = m_a {}^I\vec{V}_{a/o} = m_a \frac{1}{m_a} \cdot \sum_i m_i {}^I\vec{V}_{i/o}$$

$$\frac{d}{dt}{}^I\vec{P}_{a/o} = m_a \frac{1}{m_a} \sum_i m_i {}^I\vec{a}_{i/o} = \sum_i {}^I\vec{F}_i = \sum_i \left[{}^I\vec{F}_{i,exe} + \sum_j {}^I\vec{F}_{ij} \right] = {}^I\vec{F}_{exe}$$

Euler 2nd law (rotation): $\frac{d}{dt}{}^I\vec{h}_{s/a} = \sum_i {}^I\vec{r}_{i/a} \times {}^I\vec{F}_{i,exe} = {}^I\vec{M}_{a,exe}$

$$\begin{aligned}
\frac{d}{dt}{}^I\vec{h}_{s/a} &= \frac{d}{dt}\left[\sum_i m_i {}^I\vec{r}_{i/a} \times {}^I\vec{v}_{i/a} \right] \\
&= \sum_i m_i {}^I\vec{r}_{i/a} \times {}^I\vec{a}_{i/a} \\
&= \sum_i m_i {}^I\vec{r}_{i/a} \times \left[\vec{a}_{i/o} - {}^I\vec{a}_{a/o} \right] \\
&= \sum_i {}^I\vec{r}_{i/a} \times \vec{F}_i \\
&= \sum_i {}^I\vec{r}_{i/a} \times \left[{}^I\vec{F}_{i,exe} + \sum_j {}^I\vec{F}_{ij} \right] \\
&= \underbrace{\sum_i {}^I\vec{r}_{i/a} \times {}^I\vec{F}_{i,exe}}_{{}^I\vec{M}_{a,exe}} + \underbrace{\sum_i {}^I\vec{r}_{i/a} \times \sum_j {}^I\vec{F}_{ij}}_{0 \text{ under internal moment assumption}}
\end{aligned}$$

Newton Euler Equation: $({}^I\vec{M}_{a,exe})_{/B} = {}^B I_a \frac{d}{dt}\omega_{a/b} + \omega_{a/b} \times {}^B I_a \omega_{a/b}$

$$\begin{aligned}
{}^I\vec{M}_{a,exe} &= \frac{d}{dt}{}^I\vec{h}_{s/a} \\
&= \left[\frac{d}{dt}({}^I\vec{h}_{s/a})_{/B} \right]_{/I} + \omega_{a/I} \times {}^I\vec{h}_{s/a} \\
&= \left[\frac{d}{dt}{}^B I_a \omega_{a/b} \right]_{/I} + \omega_{a/I} \times ({}^B I_a \omega_{a/b})_{/I} \\
&= \left[{}^B I_a \frac{d}{dt}\omega_{a/b} \right]_{/I} + \left[\omega_{a/b} \times {}^B I_a \omega_{a/b} \right]_{/I} \\
&= \left[{}^B I_a \frac{d}{dt}\omega_{a/b} + \omega_{a/b} \times {}^B I_a \omega_{a/b} \right]_{/I}
\end{aligned}$$

· Newton - Euler Dynamics in SE(3)

translate in inertial frame: $m_a {}^I\vec{a}_{a/o} = \vec{F}_{exe}$

rotation in body frame: ${}^B I_a \frac{d}{dt}\omega_{a/b} + \omega_{a/b} \times {}^B I_a \omega_{a/b} = ({}^I\vec{M}_{a,exe})_{/B}$

$$x = \begin{bmatrix} {}^I\vec{r}_{a/o} \\ q_{a/I} \\ ({}^I\vec{V}_{a/o})_{/B} \\ \omega_{a/b} \end{bmatrix}$$

position, described in I frame
orientation, described in I frame
inertial velocity, described in B frame
angular velocity, described in B frame

① $\frac{d}{dt}{}^{I}\vec{r}_{a/o} = {}^{I}\vec{v}_{a/o} = q_{I\to I'} ({}^{I'}\vec{v}_{a/o})_{/B} = q_{I\to I'} * \begin{bmatrix} 0 \\ ({}^{I'}\vec{v}_{a/o})_{/B} \end{bmatrix} * q_{I\to I'}^{-1}$

② $\frac{d}{dt} q_{I\to I'} = q_{I\to I'} * \begin{bmatrix} 0 \\ \pm w_{B/I'} \end{bmatrix}$

③ $\frac{d}{dt}({}^{I}\vec{v}_{a/o})_{/I'} = \left[\frac{d}{dt}({}^{I}\vec{v}_{a/o})_{/B}\right]_{/I'} + w_{B/I'} \times {}^{I}\vec{v}_{a/o}$

$\frac{d}{dt}({}^{I}\vec{v}_{a/o})_{/B} = \left[{}^{I}\vec{a}_{a/o} - w_{B/I'} \times {}^{I}\vec{v}_{a/o}\right]_{/B}$

$= \frac{1}{m_a} {}^{B}\vec{F}_{ext} - w_{B/B} \times ({}^{I}\vec{v}_{a/o})_{/B}$

④ $\frac{d}{dt} w_{B/B} = {}^{B}I_a^{-1}\left[({}^{I}M_{a,ext})_{/B} - w_{B/B} \times {}^{B}I_a w_{B/B}\right]$

$$x = \begin{bmatrix} {}^{I}\vec{r}_{a/o} \\ q_{I\to I'} \\ ({}^{I'}\vec{v}_{a/o})_{/B} \\ w_{B/B} \end{bmatrix} \qquad \dot{x} = \begin{bmatrix} q_{I\to I'} * \begin{bmatrix} 0 \\ ({}^{I'}\vec{v}_{a/o})_{/B} \end{bmatrix} * q_{I\to I'}^{-1} \\ q_{I\to I'} * \begin{bmatrix} 0 \\ \pm w_{B/B} \end{bmatrix} \\ \frac{1}{m_a} {}^{B}\vec{F}_{ext} - w_{B/B} \times ({}^{I}\vec{v}_{a/o})_{/B} \\ {}^{B}I_a^{-1}\left[({}^{I}M_{a,ext})_{/B} - w_{B/B} \times {}^{B}I_a w_{B/B}\right] \end{bmatrix}$$